How to cite: *Angew. Chem. Int. Ed.* **2023**, *62*, e202300404
doi.org/10.1002/anie.202300404

All-Solid-State Batteries

A New Sodium Thioborate Fast Ion Conductor: $Na_3B_5S_9$

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Abstract: We report a new sodium fast-ion conductor, $Na_3B_5S_9$, that exhibits a high Na ion total conductivity of $0.80\ mS\,cm^{-1}$ (sintered pellet; cold-pressed pellet = $0.21\ mS\,cm^{-1}$). The structure consists of corner-sharing $B_{10}S_{20}$ supertetrahedral clusters, which create a framework that supports 3D Na ion diffusion channels. The Na ions are well-distributed in the channels and form a disordered sublattice spanning five Na crystallographic sites. The combination of structural elucidation via single crystal X-ray diffraction and powder synchrotron X-ray diffraction at variable temperatures, solid-state nuclear magnetic resonance spectra and ab initio molecular dynamics simulations reveal high Na-ion mobility (predicted conductivity: $0.96\ mS\,cm^{-1}$) and the nature of the 3D diffusion pathways. Notably, the Na ion sublattice orders at low temperatures, resulting in isolated Na polyhedra and thus much lower ionic conductivity. This highlights the importance of a disordered Na ion sublattice—and existence of well-connected Na ion migration pathways formed via face-sharing polyhedra—in dictating Na ion diffusion.

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Introduction

Sodium ion batteries are one of the most promising alternatives to lithium-ion batteries due to sodium's low cost, high earth abundance and widely available resources.[1–3] Similar to Li metal batteries, Na metal batteries face serious safety concerns due to the inevitable side reactions between Na metal and organic liquid electrolytes, and Na dendrite growth.[4,5] By replacing organic liquid electrolytes with solid electrolytes, all solid state sodium batteries (ASSSBs) can potentially offer better safety and achieve higher energy density.[6] Since a suitable solid electrolyte is a key component to enable the potential of ASSSBs, a significant amount of research has been expended to develop new materials in this class[7–9] that include oxides (NASICON[10,11] and β-alumina[12,13]), sulfides,[14–16] hydrides,[17] and halides.[18,19] Oxide materials including NASICON and β-alumina exhibit high total ionic conductivity of more than $1\ mS\,cm^{-1}$. A high total ionic conductivity of $5.2\ mS\,cm^{-1}$ can be achieved for a rare-earth free NASICON composition $Na_{3.4}Zr_2Si_{2.4}P_{0.6}O_{12}$.[20,21] However, oxides require high temperatures in excess of 1000 °C for synthesis and advanced pellet densification methods to achieve high ionic conductivity and minimize interfacial and grain boundary resistance. Furthermore, high temperature processing of cathode active materials and oxide solid electrolytes to form cathode composites can lead to detrimental side reactions.[22–24] Halide materials are relatively soft but exhibit much lower ionic conductivity to date ($<0.1\ mS\,cm^{-1}$), which is too low for practical applications. Hydride materials possess high ionic conductivity and ductility, but their complex synthesis requirements inhibit their practical application. On the other hand, sodium sulfide solid electrolytes exhibit high ionic conductivity—up to $41\ mS\,cm^{-1}$—comparable to that of traditional liquid electrolytes.[15,25] This has triggered significant efforts which has mainly focused on thiophosphates and their derivatives.[26] In contrast, sodium thioborate and its derivatives have been very underexplored and offer a completely different pathway towards the development of new sodium solid electrolytes.

Despite having similar material properties as thiophosphates, no sodium thioborate materials have been reported with high ionic conductivity. To date, only Na_3BS_3 and $[Na_2S]_{2/3}[(B_2S_3)_x(P_2S_5)_{1-x}]_{1/3}$ glasses have been reported with relatively low ionic conductivity ($\approx 10^{-5}\ S\,cm^{-1}$).[27,28] A few lithium thioborates with high ionic conductivity have been reported in recent years.[29–32] Theoretical work on lithium thioborate has also predicted high ionic conductivities.[33–35]

Crystalline $Li_{6+2x}B_{10}S_{18}S_x$ ($x=2$) and $Li_{7.5}B_{10}S_{18}X_{1.5}$ (X=Cl, Br, I) exhibit high ionic conductivities due to well-defined thioborate supertetrahedral networks that create void spaces which encapsulate 3D ion diffusion channels.[30,36] Moreover, supertetrahedral polyanionic networks have been reported in phosphide-type Li/Na ion conductors.[37–41] Such large 3D diffusion channels will also be beneficial for diffusion of the larger Na cation. These findings inspired us to explore sodium thioborates with a similar motif. The material $Na_3B_5S_9$ is one such candidate, whose structure published in 1996 (determined at 140 K) revealed a highly ordered Na sublattice.[42] Conductivity properties were not reported.

Here, we revisit $Na_3B_5S_9$, and show that it is the first sodium thioborate fast ion conductor to exhibit a high Na ion conductivity of 0.80 mS cm^{-1} as a sintered pellet (cold-pressed pellet = 0.21 mS cm^{-1}) and low activation energy of 0.39 eV. Our combined single crystal and powder synchrotron X-ray diffraction results resolve a highly disordered Na sublattice at room temperature in which Na ions are well-distributed in the channels created by the corner-sharing $B_{10}S_{20}$ supertetrahedral clusters. Fast Na ion conduction in $Na_3B_5S_9$ is further supported by a combination of solid-state NMR and ab initio molecular dynamics simulations.

Results and Discussion

In our synthesis, the precursors were melted and slowly cooled to favor single crystal growth. Well defined colorless plate-shaped crystals with dimensions of $100\times65\times10$ µm were picked from the as-synthesized polycrystalline agglomerate before performing powder X-ray diffraction measurements. Single crystal X-ray diffraction unequivocally confirms the structure of tetragonal $Na_3B_5S_9$ and reveals the disordered Na sublattice displayed in Figure 1. The corresponding crystallographic details are summarized in Table 1 and Tables S1, S2. The unit cell was indexed in the same tetragonal space group $I4_1/acd$ as previously reported[42] with a slightly larger unit cell $a=b=14.4406(5)$ Å, $c=26.1048(9)$ Å due to our higher measurement temperature. The structure consists of corner-sharing $B_{10}S_{20}$ supertetrahedral clusters (Figure 1a, b) that frame 3D Na-ion diffusion channels (Figure 1c). Unlike the ordered Na sublattice that forms at 140 K, a disordered Na sublattice was identified at 280 K. The originally-reported structure at 140 K consisted of three fully occupied sodium sites (Na1, Na2, Na3) that formed isolated NaS_x polyhedra connected by corner-sharing. In contrast, at 280 K, we identify all three

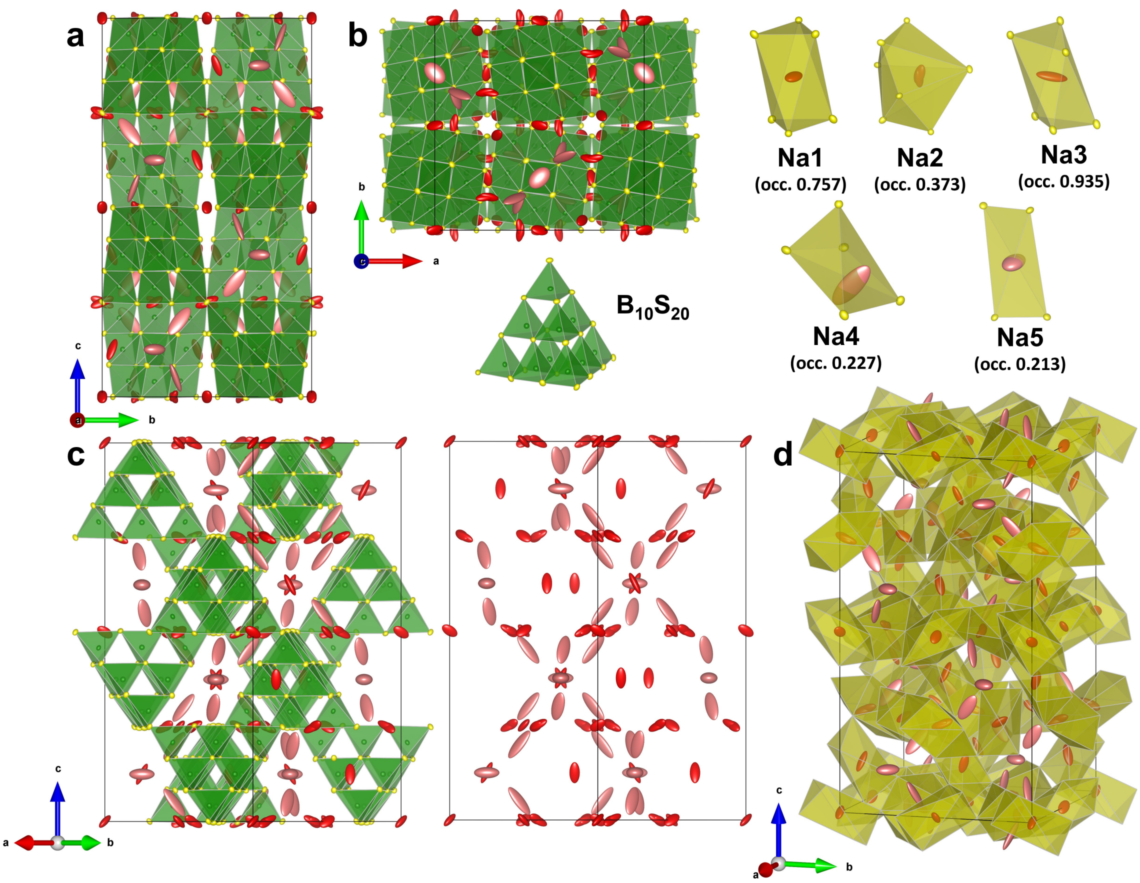

Figure 1. Structure of $Na_3B_5S_9$ from single crystal X-ray data at 280 K with all atoms shown as probability displacement ellipsoids, views along a) [100], b) [001] and c) [110] with $B_{10}S_{20}$ clusters (left) and only Na atoms (right). The structure is built from corner-sharing $B_{10}S_{20}$ clusters which form a rigid framework with large interpenetrating void spaces occupied by five disordered Na sites. d) Structure of $Na_3B_5S_9$ with only Na1, Na2, Na3 polyhedra and Na4 and Na5 ellipsoids shown: the newly identified Na4 and Na5 occupy the empty sites between Na1-3 sites and form a well-connected Na ion diffusion pathway.

Table 1: Atomic coordinates, occupation factor, and equivalent isotropic displacement parameters of $Na_3B_5S_9$ obtained from single crystal X-ray diffraction at 280 K.

Atom	Wyck. Site	*x*	*y*	*z*	Occupancy	U_{eq}
Na1	16*c*	0	0	0	0.743(7)	0.0706(12)
Na2	32*g*	0.0920(2)	0.010(3)	0.2475(9)	0.370(4)	0.073(6)
Na3	16*f*	0.2033(1)	0.4533(1)	0.125	0.932(7)	0.0923(13)
Na4	32*g*	0.1004(6)	0.1318(10)	0.1995(6)	0.235(8)	0.202(11)
Na5	8*b*	0	0.25	0.125	0.248(14)	0.157(13)
B1	32*g*	0.0237(1)	0.0981(1)	0.3749(1)	1	0.0125(3)
B2	16*d*	0	0.25	0.2900(1)	1	0.0124(5)
B3	32*g*	0.3762(1)	0.0817(1)	0.2087(1)	1	0.0110(3)
S1	32*g*	0.0901(1)	0.1842(1)	0.3315(1)	1	0.01145(9)
S2	32*g*	0.2608(1)	0.2861(1)	0.0825(1)	1	0.01434(10)
S3	32*g*	0.2826(1)	0.1394(1)	0.1656(1)	1	00144(1)
S4	32*g*	0.0821(1)	0.3135(1)	0.0034(1)	1	0.01578(10)
S5	16*e*	0.3011(1)	0	0.25	1	0.01468(10)

S.G.: $I4_1/acd$, $a=b=14.4406(5)$ Å, $c=26.1048(9)$ Å, Refined composition: $Na_{3.01}B_5S_9$.

Na1, Na2, Na3 sites with much lower occupancy. The Na2 site splits into two with a short distance of 0.32 Å, suggesting the Na^+ is delocalized over that split site and/or has high mobility. Two additional Na sites—Na4 and Na5—with lower occupancy were identified which connect the isolated Na1-3 sites and form a 3D Na ion diffusion pathway (Figure 1d). To reveal the robustness of this disordered Na ion sublattice, low temperature (200 K, instrument limited) single crystal X-ray diffraction measurements were performed on the same crystal and yielded the same disordered Na ion sublattice with some minor changes in occupancy. Crystallographic details are summarized in Tables S1, S3, S4. Maintaining the disordered Na sublattice at low temperature (200 K) indicates reasonably high bulk ionic conductivity should also be obtained.

Laboratory X-ray diffraction (Figure S1) studies prove the successful synthesis of tetragonal $Na_3B_5S_9$ as a microcrystalline powder, but due to the large unit cell of $Na_3B_5S_9$, synchrotron powder X-ray diffraction was conducted to identify the sample purity. Room temperature (300 K) high resolution synchrotron powder X-ray diffraction data were fit (Figure 2a) using the model determined by single crystal diffraction, demonstrating a phase-pure material (see Table S5 for crystallographic details). The Rietveld refinement of low temperature (140 K) synchrotron powder X-ray diffraction data (Figure 2b) confirms the Na sublattice ordering (Table S6) in agreement with the previously reported single crystal study,[42] indicating a disorder-order Na-ion sublattice transition must exist between 140 and 200 K. This temperature-driven transition demonstrates the importance of a disordered Na-ion sublattice—with abundant Na vacancies and well-connected Na-ion diffusion pathways—in dictating Na-ion conductivity. When the sites in the lattice are at similar energies and separated by low activation energy barriers, the Na-ions exhibit high mobility and migrate within the channels, leading to a disordered Na-ion sublattice.[43] At low temperature, Na-ions are frozen with significantly lower mobility and settle at low-energy crystallographic sites, giving rise to an ordered sublattice.

The Na-ion conductivity and activation energy for Na-ion transport in $Na_3B_5S_9$ were determined via temperature-dependent electrochemical impedance spectroscopy measurements on cold-pressed pellets. Room temperature Nyquist plots are shown in Figure 2c. The corresponding average room temperature Na ion conductivity is 0.21(1) $mS\,cm^{-1}$ based on three measurements to ensure reproducibility. The variable temperature (60 °C to −60 °C) Nyquist plots are shown in Figure S2. Corresponding Arrhenius plots of total ionic conductivity are shown in Figure 2d. The overall activation energy for Na ion transport determined from the Arrhenius plot is 0.39 eV. The low temperature (−20 °C to −60 °C) Nyquist plots (Figure S2d, e) clearly exhibit an asymmetrical and decompressed semicircle, which can be deconvoluted into two overlapped semicircles that represent bulk and grain boundary contributions. Due to this overlap, while the bulk and grain boundary ionic conductivity can not be accurately separated through equivalent circuit fitting, the effect of the grain boundaries is revealed by partly eliminating that contribution. Indeed, $Na_3B_5S_9$ demonstrates a nearly four-fold higher ionic conductivity of 0.80 $mS\,cm^{-1}$ upon sintering the pellet (Figure S3). This clearly indicates that the total ionic conductivity of $Na_3B_5S_9$ is governed by grain boundary resistance. A high grain boundary resistance has also been commonly observed for other sulfide materials, including $Li_7P_3S_{11}$; [44] $Li_{6+x}P_{1-x}Ge_xS_5I$; [45] $Li_{6+x}Si_xSb_{1-x}S_5I$; [46] and $Na_{3-x}(P/Sb)_{1-x}W_xS_4$.[15] Oxide materials exhibit even more significant grain boundary resistance contributions, and thus require high temperature pellet sintering or hot pressing for contact improvement and pellet densification.[20] The high grain boundary resistance is caused by poor contact between $Na_3B_5S_9$ particles within the pellet that was cold-pressed at 480 MPa for 5 mins. The SEM images (Figure 3) of the cold-pressed pellet show many void spaces in the top surface and cross-section. The corresponding density of the cold-pressed pellet estimated from its thickness, diameter and mass is only 84 % of theoretical density. Even the sintered pellet density still only reaches 87 % (Figure S4). In principle, hot pressing the material would better densify the pellets, and

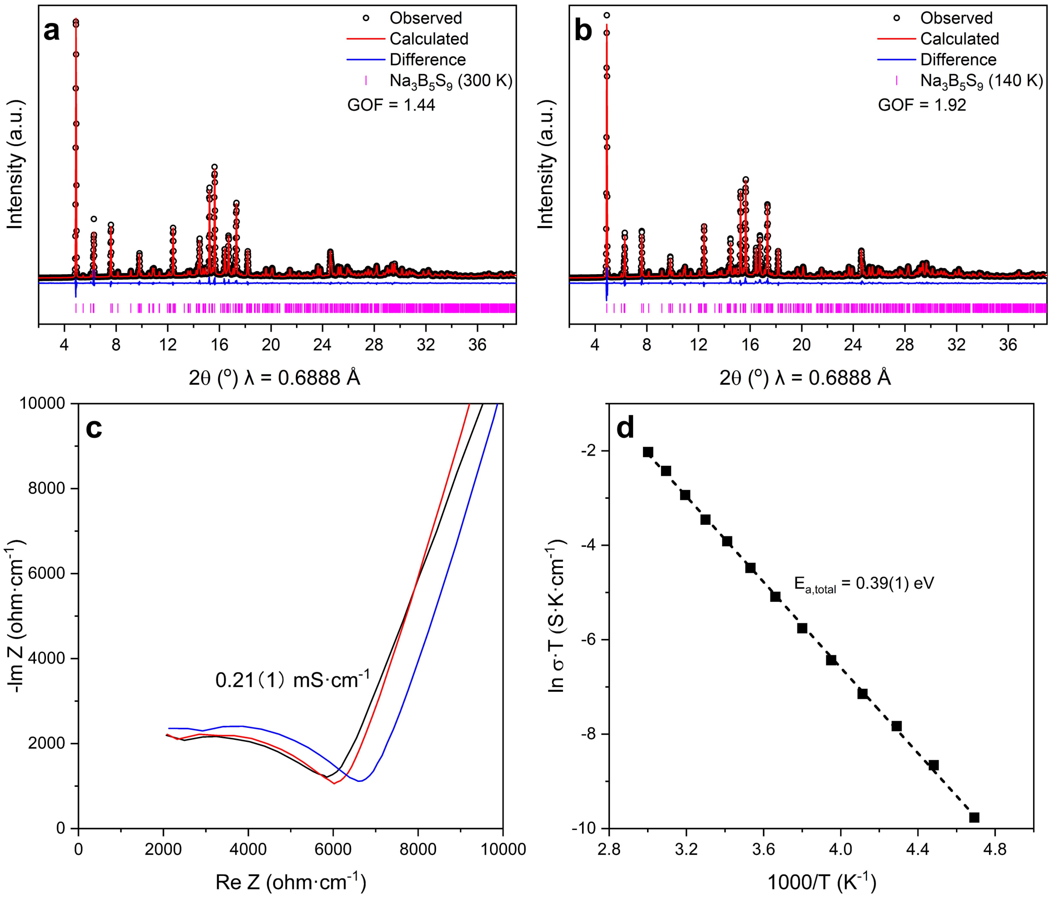

Figure 2. Synchrotron diffraction patterns and the corresponding Rietveld refinement fits of $Na_3B_5S_9$ at a) 300 K and b) 140 K. Experimental data are shown in black circles; the red line denotes the calculated patterns; the difference profile is shown in blue, and the calculated positions of the Bragg reflections are shown as vertical ticks. c) Nyquist plots of $Na_3B_5S_9$ for three different batches with average ionic conductivity of 0.21 mS cm^{-1} at 250 MPa. d) Arrhenius plots of the conductivity values for $Na_3B_5S_9$ in the temperature range from 60 to −60 °C.

attain closer to the intrinsic bulk ionic conductivity value but this would require operation under controlled atmosphere conditions.

Furthermore, the voids on the pressed pellet surface (Figure 3b) will promote metal dendrite growth.[47] The corresponding Na metal symmetric cell exhibits short cycle life at relatively low current density and areal capacity (Figure S5). $Na_3B_5S_9$ is also not thermodynamically stable with Na metal, as B^{3+} can be easily reduced to form Na–B alloy, as previously observed for Na_3BS_3 glass.[48] The XPS of $Na_3B_5S_9$ powder mixed with Na metal (Figure S6) clearly shows the formation of Na_2S and Na–B alloy. The interfacial resistance growth was probed by impedance measurements on a Na|$Na_3B_5S_9$|Na symmetric cell. The interfacial resistance (Figure S7) increased over time which corresponds to the growth of non-conductive Na_2S as an interphase. Linear sweep voltammetry measurements and DFT calculations (Figure S8) were performed to assess its electrochemical stability voltage window, which is relatively narrow. It lies between 1.2 V–2.2 V vs Na^+/Na, similar to other sulfide Na solid electrolytes.[49] Such a narrow voltage window makes it incompatible with commonly-used sodium cathode active materials at high voltage unless coatings are applied to the cathode. Thus, an ASSSB with a low voltage Chevrel-phase Mo_6S_8 cathode and $Na_{15}Sn_4$ anode was constructed to demonstrate the function of $Na_3B_5S_9$ as a sodium-ion solid electrolyte (Figure S9), which achieved close to theoretical capacity of Mo_6S_8. Oxygen-substitution has been shown to widen the electrochemical window in lithium oxythioborosilicate glass solid electrolytes,[29] a concept which in principle can also be applied to $Na_3B_5S_9$. Cathode surface coating with an ionic conductive and electronic insulting interface can also be applied to stabilize the interface between $Na_3B_5S_9$ and cathode active materials. This strategy has been applied in solid state Li batteries but is beyond the scope of this work.[50]

^{11}B and ^{23}Na magic-angle-spinning (MAS) solid-state NMR spectra obtained on $Na_3B_5S_9$ at 304 K using Hahn echo acquisitions are presented in Figure 4a, b. The ^{11}B spectrum (Figure 4a) is dominated by an asymmetric BS_4 isotropic peak, with tiny additional signals corresponding to trigonal planar ^{11}B sites, both BS_3 (minor Na_3BS_3 impurity, ≈3.3 % of the overall signal) and BO_3 (minor Na_3BO_3 impurity, ≈1.9 %), as previous work suggested.[51–53] Examination of the sideband manifold reveals that the asymmetry of the BS_4 peak is indeed due to two distinct sites, as

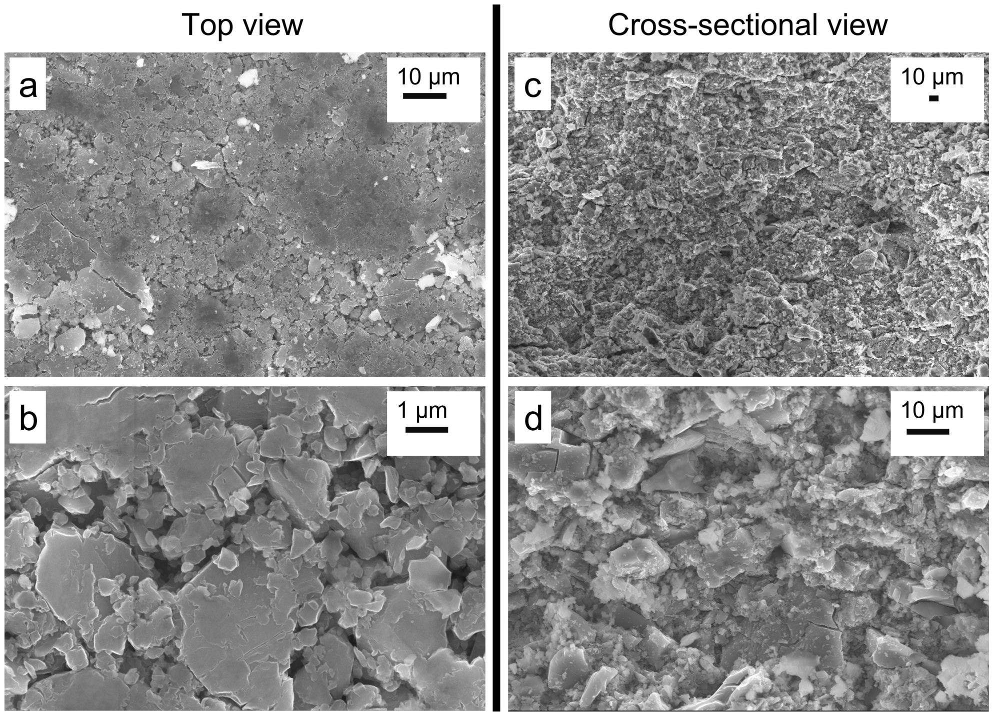

Figure 3. SEM images of a $Na_3B_5S_9$ pellet cold-pressed at 480 MPa for 5 mins; a), b) top view and c), d) cross-sectional view. Images show a low pellet density and the presence of voids.

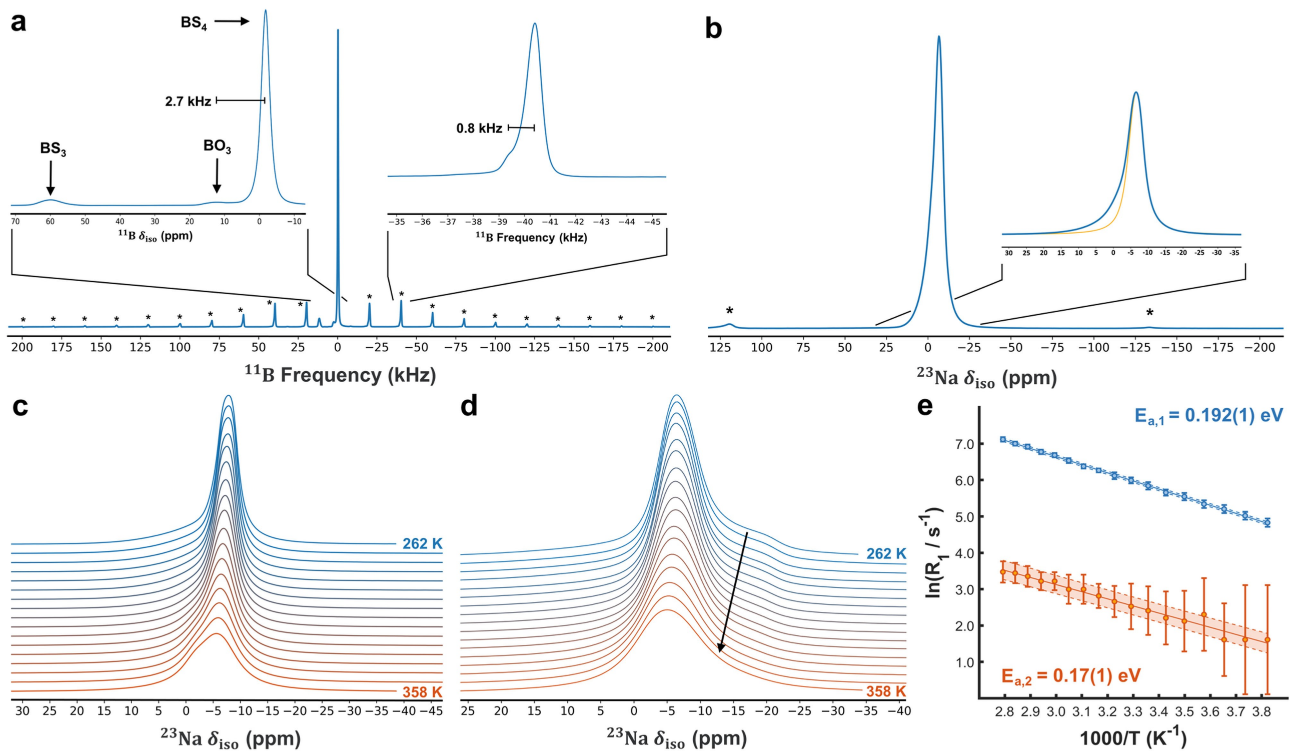

Figure 4. NMR studies of $Na_3B_5S_9$ with a) ^{11}B 20 kHz MAS Hahn echo and b) ^{23}Na 20 kHz MAS at 304 K and 14.1 T; The single Gaussian fit component in the inset is presented purely for illustrative purposes to clarify the high-frequency skew of the isotropic. Variable-temperature ^{23}Na MAS c) and static d) line shape evolution demonstrates a complex evolution of the ion site exchange dynamics. e) Arrhenius plot of ^{23}Na T_1 relaxation measurements, which exhibit bi-exponential behavior with low activation energies.

opposed to chemical shift dispersion (i.e. disorder), albeit with a high degree of signal overlap. Nevertheless, the shift separation of ≈ 0.8 kHz in the sideband splitting clearly demonstrates that this second peak is distinct from the nearby BO_3 impurity signal (≈ 2.7 kHz from the primary BS_4 isotropic). A simultaneous deconvolution of the ^{11}B static line shapes acquired with a Hahn echo and a composite-pulse approach (Figure S10) reveals approximate proportions of 0.41 and 0.59 for the two BS_4 signals, consistent with the expected signal fractions for corner *vs* edge species of BS_4 tetrahedra (4:6) in the $B_{10}S_{20}$ supertetrahedral clusters. The resultant line shape parameters are summarized in Table S7. Interestingly, the corner tetrahedra exhibit a much smaller C_Q and asymmetry parameter than the edge sites, implying association of Na-ions with corner BS_4 tetrahedral faces of the supertetrahedra, yielding an enhanced electric field gradient in these locations.

The ^{23}Na MAS in Figure 4b at 304 K exhibits a lone, asymmetric isotropic peak sloping slightly to high frequency, with the comparatively small sideband manifold and narrowness of the feature already indicative of rapid motion amongst the Na sites. As is well known, rapid site exchange relative to the *O*(ms) NMR time scale results in coalesced peaks,[54] which are the weighted average of the NMR shifts and peak areas in the absence of exchange. Data from variable-temperature (VT) experiments, however—both static and under MAS—are considerably more complicated (Figure 4c, d). At low temperature, the narrow central transition persists (Figure 4c), indicating some sites are still in rapid exchange even at 255 K, and the high-frequency sloping feature becomes more pronounced. This is accompanied by a pronounced quadrupolar line shape in the static spectrum in Figure 4d superimposed on a comparatively narrow Lorentzian line shape. The quadrupolar contribution may be ascribed to the Na3 site, based upon its expected signal fraction (Table S8). It is also potentially the comparatively less-mobile site responsible for the large ^{11}B "supertetrahedron edge site" C_Q.

Conversely, at high temperature (348 K), a second narrow signal (Figure 4c) emerges at slightly higher frequency than the main signal in the ^{23}Na MAS spectrum (having transitioned via the intermediate line shape in Figure 4b). However, the static ^{23}Na spectrum (Figure 4d) at the same temperature gives a broad Lorentzian-like line shape, indicative of a coalescence of the previously distinct quadrupolar and Lorentzian line shapes at that existed for 262 K static. Such coalescence behaviour has previously been observed in Na-ion conductors/ionomers.[55,56] Typically this occurs over a much larger temperature range because the enhanced dynamics emerge more gradually owing to low Na^+ lattice mobility. However, a correspondence between the signal fractions and the crystallographic sites is difficult to fully establish because of the exchange effects. The ^{11}B line shape characteristics and signal fractions are not significantly impacted over this same temperature range, consistent with B–S moieties forming the rigid tunnel framework (Figure S11).

To further elucidate and quantify the Na^+ dynamics, ^{23}Na saturation recovery measurements (static) of the longitudinal relaxation rate, T_1, were performed as function of temperature. NMR relaxation is induced by fluctuating short-range interactions, and as such is a measure of local motional dynamics in materials such as ion conductors.[57] Integration of the full line shape as a function of the recovery time yielded a bi-exponential time-dependence. While multiple sites are observed from the diffraction analysis, the explicit deconvolution of the multi-component behavior in the relaxation curve to site-specific attributions is not attempted here and will be the subject of a future investigation. As a result, a direct bi-exponential ("model-free") fit of the total integrated intensity was performed (Figure 4e), split into "large" and "small" signal components, to simply extract the temperature dependence and resulting activation energies, with the two signals giving respective values of 0.192(1) eV (large signal) and 0.17 (1) eV (small signal). These compare favorably with other Na-ion conductors which have been previously investigated by ^{23}Na relaxation measurements (0.19 eV for NASICON-type $Na_{3.4}Sc_2(SiO_4)_{0.4}(PO_4)_{2.6}$;[58] 0.19 eV and 0.21 eV for cubic and tetragonal Na_3PS_4 (respectively);[59] 0.29 eV for sodium strontium silicate[60]). This is a strong metric for facile local motion, but it is again challenging to link it directly with dynamics of individual Na site populations as extracted from the NMR line shapes, without invoking additional assumptions regarding how the slice-by-slice signal attribution is performed. The site exchange-rate time scale concomitant with the recovery time scale competes with relaxation and the residual influence of incompletely-averaged satellite transitions.[61,62] This latter effect also produces bi-exponential behaviour for a $S=3/2$ nucleus—interestingly, in the approximate proportions[63,64] of 4/5 to 1/5 resulting from the model-free, two-component T_1 fit. Further exploration of these complex dynamical processes by magnetic resonance methods will be the subject of a future study.

Na-ion diffusion in $Na_3B_5S_9$ was also explored with AIMD simulations using the VASP platform and starting from the refined single crystal structure shown in Figure 1. The Na-ion probability density isosurface from AIMD simulation at 1000 K is shown in Figure 5a and clearly demonstrates the 3D Na-ion diffusion pathways and significant Na-ion mobility. All the Na atoms in the five distinct crystallographic sites take part in the ion diffusion, owing to the rich vacancy population. The Na-ion diffusivity was calculated from the mean square displacement of the Na atoms from AIMD simulations. The Arrhenius plot generated from of self-diffusion coefficients calculated at temperatures from 800 K to 1100 K is shown in Figure 5b. The extrapolated room temperature Na-ion self-diffusivity is $1.746\times10^{-8}\,cm^2\,s^{-1}$ which corresponds to a theoretical room temperature conductivity of $0.96\,mS\,cm^{-1}$ and a fitted activation energy for Na-ion diffusion is 0.28(2) eV. The theoretical room temperature conductivity value is in well agreement with the measured sintered pellet ionic conductivity.

Although oxide materials i.e., NaSICON[10,11] and β-alumina[12,13] exhibit higher ionic conductivity and wider electrochemical stability windows than $Na_3B_5S_9$, their syn-

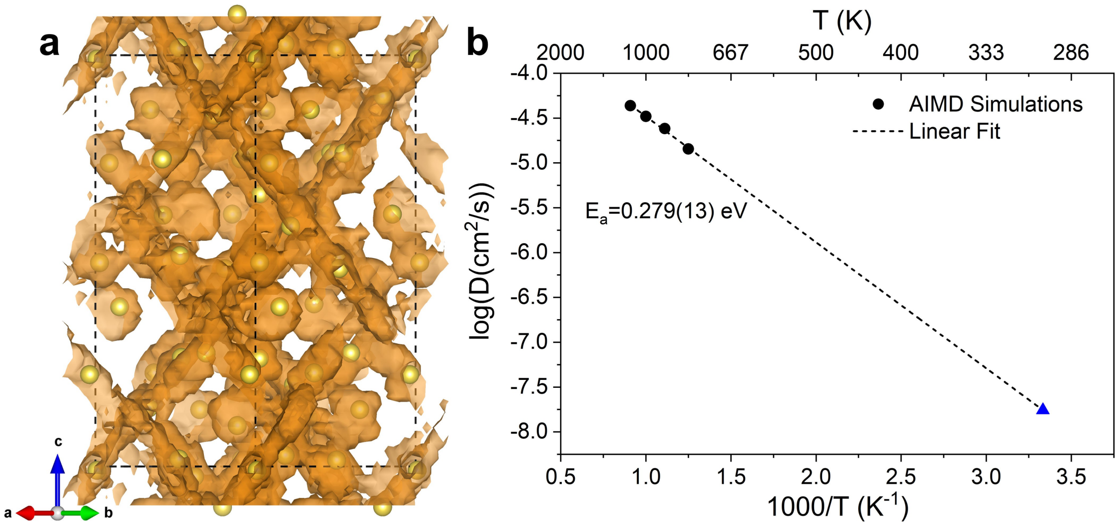

Figure 5. a) Na-ion probability density isosurface (orange) plot obtained from ab initio molecular dynamics (AIMD) studies at 1000 K for ≈40 ps, demonstrating the 3D Na-ion diffusion pathway: b) Arrhenius plot of Na-ion diffusivity and theoretical ionic conductivity in $Na_3B_5S_9$ from AIMD simulations.

thesis and processing require significantly higher temperature and precise composition control, which significantly increases their cost. Furthermore, due to the high hardness of oxides, it is difficult for oxide solid electrolytes to achieve good contact with most cathode active materials in cathode composites, which restricts their use and/or requires incorporation of polymer solid electrolytes/ionic liquids.[65] In contrary, sulfide electrolytes are much softer than oxides, though some grain boundary resistance can still be observed (significantly less detrimental than oxides). Thus, cathode composites with sulfide solid electrolytes are easier to process. At present, while $Na_3B_5S_9$ exhibits lower ionic conductivity compared to NaSICON, it represents the first fast ion conductor in the low-cost, low-mass thioborate material class and opens up a new direction for the exploration of Na solid state electrolyte materials.

Conclusion

We find that $Na_3B_5S_9$ is a fast ion conductor at room temperature which exhibits a corner-sharing $B_{10}S_{20}$ supertetrahedral framework and a Na-ion sublattice disordered over 5 distinct crystallographic sites. Single crystal X-ray diffraction combined with synchrotron X-ray powder diffraction studies allow the precise determination of this disordered Na-ion sublattice at room temperature that is maintained down to 200 K and confirm the existence of Na-ion sublattice ordering at even lower temperatures. The disordered Na-ion sublattice leads to high Na-ion conductivity, whereas the disorder-order transition at low temperature (between 140 K and 200 K) triggers formation of an ordered Na sublattice with low ionic conductivity. While $Na_3B_5S_9$ exhibits a room temperature total ionic conductivity of 0.21 mS cm^{-1} and a relatively low activation energy of 0.39 eV for cold-pressed pellet, the sintered pellet total ionic conductivity reaches 0.80 mS cm^{-1}. The high ionic conductivity was further supported by solid state ^{23}Na NMR spectroscopy and theoretical studies via AIMD simulations. ^{23}Na NMR reveals fast Na-ion site exchange amongst the sites which indicates facile Na-ion migration within the lattice. AIMD simulation confirms a disordered Na-ion sublattice rich in vacancies, and high in Na-ion mobility that provides a 3D Na-ion diffusion pathway with relatively low activation energy (0.96 mS cm^{-1}, 0.28 eV).

Although $Na_3B_5S_9$ exhibits a narrow electrochemical window, tweaking that window via substitution i.e., synthesis of $Na_3B_5S_{9-x}O_x$ should improve its electrochemical performance, as demonstrated in other solid electrolytes.[29,66,67] Furthermore, the ionic conductivity of $Na_3B_5S_9$ could be further improved by increasing the Na-ion concentration, i.e., by incorporation of NaX (X=Cl, Br, I) in the large channel created by the corner-sharing $B_{10}S_{20}$ supertetrahedral network, similar to the Li thioborate halide.[30] Such studies are underway in our lab and will be the subject of further reports.

Supporting Information

Experimental methods, details of AIMD simulations, and additional diffraction, electrochemistry, XPS, and NMR data, are given in the Supporting Information. Deposition Numbers 2235210 and 2235211 contain the supplementary crystallographic data for this paper. These data are provided free of charge by the joint Cambridge Crystallographic Data Centre and Fachinformationszentrum Karlsruhe Access Structures service.

Author Contributions

L.Z. conceived and designed the experimental work. L.Z. performed the synthesis of the solid electrolytes, powder X-ray diffraction measurements, structural resolution of powder synchrotron diffraction and the electrochemical measurements. J.D.B. performed the NMR measurements and data analysis. B.S. performed the theoretical simulations. C.L. performed the SEM measurements and XPS data analysis. A.A. performed single-crystal diffraction and structure resolution. L.Z., J.D.B. and L.F.N. wrote the manuscript with input from all authors.

Acknowledgements

This work was fully supported by the Joint Center for Energy Storage Research, an Energy Innovation Hub funded by the US Department of Energy, Office of Science, Basic Energy Sciences. Underpinning resources were provided from NSERC via their Discovery Grant program, and the Ontario Research Foundation through their Excellence Fund. Synchrotron powder XRD measurements were performed using CMCF-BM beamline at the Canadian Light Source, a national research facility of the University of Saskatchewan, which is supported by the Canada Foundation for Innovation (CFI), the Natural Sciences and Engineering Research Council (NSERC), the National Research Council (NRC), the Canadian Institutes of Health Research (CIHR), the Government of Saskatchewan, and the University of Saskatchewan. We thank Dr. Joel Reid for the synchrotron X-ray powder diffraction data collection, Dr. Peter Broderson (University of Toronto) for carrying out the XPS experiments.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the Supporting Information of this article.

Keywords: All Solid State Sodium Batteries · Disordered Na-Ion Sublattice · Sodium Fast-Ion Conductors · Supertetrahedral Cluster · Thioborate

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Manuscript received: February 7, 2023
Accepted manuscript online: May 17, 2023
Version of record online: June 20, 2023